

DevOps Assignment Submission

1. Backup Script (web_backup.sh)

```
#!/bin/bash
```

```
DATE=$(date +%F)
```

```
BACKUP_DIR="/backups"
```

```
LOG_FILE="/var/log/web_backup.log"
```

```
# Apache Backup
```

```
tar -czf $BACKUP_DIR/apache_backup_$DATE.tar.gz /etc/httpd /var/www/html
```

```
# Verify Apache Backup
```

```
tar -tzf $BACKUP_DIR/apache_backup_$DATE.tar.gz > /dev/null && echo "Apache backup  
verified on $DATE" >> $LOG_FILE
```

```
# Nginx Backup
```

```
tar -czf $BACKUP_DIR/nginx_backup_$DATE.tar.gz /etc/nginx /usr/share/nginx/html
```

```
# Verify Nginx Backup
```

```
tar -tzf $BACKUP_DIR/nginx_backup_$DATE.tar.gz > /dev/null && echo "Nginx backup  
verified on $DATE" >> $LOG_FILE
```

2. Required Screenshots

- htop command output

```
0% 4[
1% 5[
2% 6[
3% 7[
Mem[|||||] 351M/7.61G Tasks: 38, 10 thr, 0 kthr; 1 running
Swap[|] 0K/2.00G Load average: 0.03 0.01 0.00
Uptime: 00:11:32

Main | /top
PID USER      PRI  NI  VIRT   RES   SHR  S  CPU% MEM%   TIME+  Command
1 root        20    0  22216 12720 9264 S  0.0  0.2  0:00.09 /sbin/init
2 root        20    0  3120  2048  2048 S  0.0  0.0  0:00.00 /init
6 root        20    0  3120  1792  1792 S  0.0  0.0  0:00.00 |
7 root        20    0  3120  1792  1792 S  0.0  0.0  0:00.00 |
8 root        20    0  3120  2048  2048 S  0.0  0.0  0:00.00 |
322 root       20    0  3124   900   768 S  0.0  0.0  0:00.00 |
323 root       20    0  3140  1160  1024 S  0.0  0.0  0:00.02 |
324 administra 20    0  6072  5248  3584 S  0.0  0.1  0:00.01 |
490 root       20    0 14316  6996  5760 S  0.0  0.1  0:00.06 |
491 root       20    0 14316  2536  1280 S  0.0  0.0  0:00.00 |
492 root       20    0  6800  4096  3712 S  0.0  0.1  0:00.00 |
516 root       20    0  5016  4096  3456 S  0.0  0.1  0:00.00 |
1565 root      20    0  6584  5248  3456 R  0.7  0.1  0:01.19 |
325 root       20    0  6664  4224  3712 S  0.0  0.1  0:00.01 |
396 administra 20    0  6072  5120  3584 S  0.0  0.1  0:00.01 |
39 root       19   -1 50444 15360 14592 S  0.0  0.2  0:00.18 |
80 root       20    0 25144  6272  4992 S  0.0  0.1  0:00.10 |
141 systemd-re 20    0 21460 12544 10368 S  0.0  0.2  0:00.09 |
143 systemd-ti 20    0 91028  7552  6784 S  0.0  0.1  0:00.06 |
152 systemd-ti 20    0 91028  7552  6784 S  0.0  0.1  0:00.00 |
157 root       20    0  4236  2432  2304 S  0.0  0.0  0:00.00 |
158 messagebus 20    0  9888  4992  4352 S  0.0  0.1  0:00.15 |
170 root       20    0 18176  8064  7296 S  0.0  0.1  0:00.07 |
177 root       20    0  3160  1920  1792 S  0.0  0.0  0:00.00 |
181 syslog     20    0 217M  5376  4224 S  0.0  0.1  0:00.03 |
217 syslog     20    0 217M  5376  4224 S  0.0  0.1  0:00.01 |
218 syslog     20    0 217M  5376  4224 S  0.0  0.1  0:00.00 |
219 syslog     20    0 217M  5376  4224 S  0.0  0.1  0:00.00 |
184 root       20    0  3116  1664  1664 S  0.0  0.0  0:00.00 |
190 root       20    0 11164  1720   768 S  0.0  0.0  0:00.00 |
191 www-data    20    0 12888  4152  2944 S  0.0  0.1  0:00.00 |
192 www-data    20    0 12888  4152  2944 S  0.0  0.1  0:00.00 |
193 www-data    20    0 12888  4152  2944 S  0.0  0.1  0:00.00 |
194 www-data    20    0 12888  4408  3072 S  0.0  0.1  0:00.00 |
195 www-data    20    0 12888  4280  3072 S  0.0  0.1  0:00.00 |
196 www-data    20    0 12888  4280  3072 S  0.0  0.1  0:00.00 |
197 www-data    20    0 12888  4152  2944 S  0.0  0.1  0:00.00 |
198 www-data    20    0 12888  4152  2944 S  0.0  0.1  0:00.00 |
205 root       20    0 104M 22016 12928 S  0.0  0.3  0:00.12 |
256 root       20    0 104M 22016 12928 S  0.0  0.3  0:00.00 |
371 administra 20    0 20308 11136 9088 S  0.0  0.1  0:00.07 |
372 administra 20    0 21160  3520  1792 S  0.0  0.0  0:00.00 |
495 root       20    0 20316 11136 9088 S  0.0  0.1  0:00.07 |
496 root       20    0 21168  3388  1664 S  0.0  0.0  0:00.00 |
783 polkitd    20    0 300M  7680  6912 S  0.0  0.1  0:00.02 |
785 polkitd    20    0 300M  7680  6912 S  0.0  0.1  0:00.00 |
786 polkitd    20    0 300M  7680  6912 S  0.0  0.1  0:00.00 |
787 polkitd    20    0 300M  7680  6912 S  0.0  0.1  0:00.00 |

Help F2Setup F3Search F4Filter F5List F6Sortby F7Nice F8Nice F9Kill F10Quit
```

- df -h disk usage

```
root@DESKTOP-6R376G6:/home/administrator# df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G  0  3.9G  0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            3.9G  4.0K  3.9G  1% /mnt/wsl
drivers         477G  354G  124G  75% /usr/lib/wsl/drivers
/dev/sdd        1007G  2.4G  954G  1% /
none            3.9G  76K  3.9G  1% /mnt/wslg
none            3.9G  0  3.9G  0% /usr/lib/wsl/lib
rootfs          3.8G  2.7M  3.8G  1% /init
none            3.9G  508K  3.9G  1% /run
none            3.9G  0  3.9G  0% /run/lock
none            3.9G  0  3.9G  0% /run/shm
none            3.9G  76K  3.9G  1% /mnt/wslg/versions.txt
none            3.9G  76K  3.9G  1% /mnt/wslg/doc
C:\             477G  354G  124G  75% /mnt/c
tmpfs           779M  20K  779M  1% /run/user/1000
tmpfs           779M  20K  779M  1% /run/user/0
root@DESKTOP-6R376G6:/home/administrator#
```

- User creation (sarah & mike)

```
root@DESKTOP-6R376G6:/home/administrator# sudo adduser sarah
adduser mikesudo adduser mikeinfo: Adding user `sarah' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `sarah' (1001) ...
info: Adding new user `sarah' (1001) with group `sarah (1001)' ...
info: Creating home directory `/home/sarah' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for sarah
Enter the new value, or press ENTER for the default
    Full Name []: sarah
    Room Number []: 01
    Work Phone []:
    Home Phone []:
    Other []:
```

```
root@DESKTOP-6R376G6:/home/administrator# sudo adduser mike
info: Adding user `mike' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `mike' (1002) ...
info: Adding new user `mike' (1002) with group `mike (1002)' ...
info: Creating home directory `/home/mike' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for mike
Enter the new value, or press ENTER for the default
    Full Name []: mike
    Room Number []:
    Work Phone []:
```

- Password policy (chage -l sarah)

```
sudo chmod 700 /home/mike
root@DESKTOP-6R376G6:/home/administrator# sudo chage -M 90 -m 7 -W 7 mike
root@DESKTOP-6R376G6:/home/administrator# chage -l mike
Last password change                : Mar 19, 2026
Password expires                    : Jun 17, 2026
Password inactive                   : never
Account expires                     : never
Minimum number of days between password change : 7
Maximum number of days between password change : 90
Number of days of warning before password expires : 7
```

- Backup files (ls /backups)

```
root@DESKTOP-6R376G6:/home/administrator# sudo mkdir -p /backup
root@DESKTOP-6R376G6:/home/administrator# sudo chmod 755 /backup
root@DESKTOP-6R376G6:/home/administrator# sudo nano /usr/local/bin/backup.sh
root@DESKTOP-6R376G6:/home/administrator# sudo chmod +x /usr/local/bin/backup.sh
root@DESKTOP-6R376G6:/home/administrator# ls /backup
root@DESKTOP-6R376G6:/home/administrator# sudo chmod 755 /backup
root@DESKTOP-6R376G6:/home/administrator# sudo /usr/local/bin/backup.sh
tar: Removing leading '/' from member names
root@DESKTOP-6R376G6:/home/administrator# ls /backup
home_backup_2026-03-19.tar.gz
root@DESKTOP-6R376G6:/home/administrator# sudo nano /usr/local/bin/backup.sh
root@DESKTOP-6R376G6:/home/administrator# sudo chmod +x /usr/local/bin/backup.sh
root@DESKTOP-6R376G6:/home/administrator# sudo mkdir -p /backups
root@DESKTOP-6R376G6:/home/administrator# sudo chmod 755 /backups
root@DESKTOP-6R376G6:/home/administrator# sudo mkdir -p /backups
root@DESKTOP-6R376G6:/home/administrator# sudo chmod 755 /backups
root@DESKTOP-6R376G6:/home/administrator# sudo nano /usr/local/bin/web_backup.sh
root@DESKTOP-6R376G6:/home/administrator# sudo chmod +x /usr/local/bin/web_backup.sh
root@DESKTOP-6R376G6:/home/administrator# sudo /usr/local/bin/web_backup.sh
```

- Backup logs (cat /var/log/web_backup.log)

```
root@DESKTOP-6R376G6:/home/administrator# cat /var/log/web_backup.log
Apache backup verified on 2026-03-20
Nginx backup verified on 2026-03-20
root@DESKTOP-6R376G6:/home/administrator# crontab -e
```

- Cron job (crontab -l)

```
root@DESKTOP-6R376G6:/home/administrator# (crontab -l 2>/dev/null; echo "0 0 * * 2 /usr/local/bin/web_backup.sh") | crontab -
root@DESKTOP-6R376G6:/home/administrator# crontab -l
0 0 * * 2 /usr/local/bin/web_backup.sh
root@DESKTOP-6R376G6:/home/administrator# htop
root@DESKTOP-6R376G6:/home/administrator#
```

3. Report Summary

System monitoring, user management, and automated backup setup were successfully implemented.